Meet Solanki

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Education

Bachelor of Science (Honours), Computer Science – Software Engineering *Jan 2023 – Expected Apr 2027*
University of Windsor *Windsor, ON*

- **GPA**: 3.7 / 4.0
- **Dean's Honour Roll:** 2023–2024, 2024–2025
- **Coursework:** DBMS, ML, AI, DSA, Cloud Computing, Operating Systems, System Programming, Web Development, Big Data Analysis, Advanced DB, Software development, Software Architecture

Technical Skills

Programming Languages: Python, Java, C/C++, C/.Net, React, PHP, JavaScript, R, SQL, Bash
AI & Machine Learning: PyTorch, Hugging Face Transformers (LLMs, NLP), Scikit-learn, YOLOv8, CNNs, OpenCV, Grad-CAM, NER, Attention Models, Optimization, Forecasting
Data Engineering & Analytics: Pandas, NumPy, Power BI, Matplotlib, Seaborn, ETL Automation, Data Cleaning, Data Pipelines
Software Development: FastAPI, Streamlit, REST APIs, Object-Oriented Design, Version Control (Git), Testing & Debugging
Database Systems: MySQL, Oracle DB, MSSQL, MongoDB (NoSQL), Snowflake, Schema Design, Query Optimization
Cloud & DevOps: AWS, Azure, Docker, CI/CD Pipelines, GitHub Actions, MLOps, Containerization
Development Tools: GitHub, VS Code, PyCharm, Azure DevOps, Postman

Work Experience

Software Developer (Co-op) – Centerline *Jan 2026 – Present*
C#, .NET, React, Azure ML, Event Hubs, Azure SQL, Cosmos DB, ADX, KQL, Redis *Windsor, ON*

- Develop full-stack features for an Azure-based IoT analytics web application.
- Build custom C#/.NET APIs, React web apps, Azure Function Apps, and business logic libraries with Redis caching to optimize data queries.
- Process and analyze IoT telemetry using Event Hubs, Azure Data Explorer, and KQL.
- Integrate Azure SQL, Cosmos DB, and ADX across relational, NoSQL, and analytical workloads.
- Train anomaly-detection models in Azure ML on 100,000+ IoT records, achieving 92% accuracy.
- Follow software architecture standards and industry best practices across development, testing, and production environments.
- Support secure deployments using Azure DevOps, CI/CD, VNets, and private endpoints.

Research Database Programmer – University of Windsor *Sep 2025 – Dec 2025*
MySQL, Oracle DBMS, Python, PHP, WEKA *Windsor, ON*

- Develop research-oriented database tools and scripts using MySQL, Oracle, Python, and PHP.
- Set up controlled test environments to evaluate query execution and indexing performance.
- Create preprocessing and data-mining workflows with SQL, Python, and WEKA for analysis.
- Support MSc/PhD researchers by dry running algorithms and validating experimental outcomes.
- Document experiment setups, findings, and procedures to ensure reproducible research workflows.

Artificial Intelligence Intern – Glendor INC *May 2025 – Aug 2025*
Python, PyTorch, Transformers, OpenCV, NLP, NER *United States • Remote*

- Developed deep learning pipelines for facial and mask detection using CNNs (PyTorch, OpenCV), achieving ∼95% accuracy on large datasets.
- Built an NLP/LLM pipeline for PHI extraction and redaction, achieving 98% precision/recall and improving compliance for healthcare clients.
- Fine-tuned Hugging Face Transformer models to classify sensitive health text, improving data-security workflows and model robustness.
- Collaborated on real-time visual object extraction methods, enhancing processing efficiency for production deployment.

Data Specialist Assistant (Co-op) – South Essex Community Council *Jan 2025 – Apr 2025*
Power BI, MySQL, PHP, JavaScript, CRM Systems *Leamington, ON*

- Built and optimized CRM modules in PHP and MySQL, reducing report generation time by 40%.

- Automated ETL workflows with Power BI and Power Query, saving 15+ hours per month for multiple departments.
- Designed interactive Power BI dashboards used by leadership for operational decision-making.
- Implemented OTP authentication and encryption for secure cloud-based CRM access.
- Developed an internal address search tool replacing third-party APIs, reducing operational costs.

Teaching Assistant – University of Windsor | *Sept 2024 – Dec 2025*

C, Python, SQL, PyTorch, UNIX, Web Development, OOP | *Windsor, ON*

- Supported instruction across five courses: COMP2560 (Systems Programming), COMP3220 (OOSAD), COMP3150 (Database Systems), COMP3340 (Web Development), and COMP3710 (Artificial Intelligence).
- Delivered lab sessions for 40+ students, teaching C in UNIX, SQL/schema design, OOP principles, UML modeling, and full-stack web fundamentals (HTML, CSS, JS, PHP).
- Assisted students with AI fundamentals including search algorithms, Python ML workflows, and introductory PyTorch model development.
- Graded 200+ assignments, tests, and projects with consistency while ensuring academic integrity.
- Contributed to faculty research initiatives through experimentation, debugging, and technical analysis.

Academic Projects

TrafficIQ – Intelligent Traffic Forecasting & Analytics (In Progress) | *Aug 2025 – Nov 2025*

Technologies: Kafka, Spark Streaming, Python, Scikit-learn, Power BI

- Designing a Kafka + Spark pipeline to stream and process live traffic data for real-time analytics.
- Developing predictive models in Scikit-learn to forecast congestion patterns with a target accuracy of $\sim$80%.
- Building Power BI dashboards to visualize heatmaps and KPIs for urban traffic planning.

MediScan – AI Powered Bone Fracture Detection | *May 2025 – Jul 2025*

Technologies: Python, PyTorch, YOLOv8, FastAPI, OpenCV, Grad-CAM

- Built an AI imaging model with YOLOv8 to detect multiple fracture types from X-rays in under 2s, achieving $\sim$93% accuracy.
- Applied Grad-CAM to generate explainability heatmaps, improving model interpretability for clinicians.
- Deployed inference pipeline with FastAPI and REST APIs to deliver real-time results via a web interface.
- Fine-tuned PyTorch models on the FracAtlas dataset to improve robustness across diverse imaging conditions.

Bilingo – AI Translator (English–French) | *Jan 2025 – Apr 2025*

Technologies: Python, Hugging Face Transformers (NLLB-200), Streamlit

- Built a bilingual NLP translation system using NLLB-200, achieving $>$90% accuracy on English–French test phrases.
- Developed a lightweight Streamlit UI to deliver instant browser-based translations for non-technical users.
- Integrated pre-trained Hugging Face models to accelerate deployment and reduce training overhead.
- Optimized inference for GPU execution, reducing translation latency to under 1s per sentence.

KnowAI – LLM Prototype for Detecting Outdated Answers | *Dec 2024*

Technologies: Python, Hugging Face Transformers (BERT, RoBERTa, DistilBERT), Scikit-learn, Pandas

- Defined the problem of outdated Stack Overflow answers arising from deprecated APIs and evolving libraries.
- Curated and labeled a dataset of 1,200+ answers to train and evaluate classification models.
- Applied pre-trained LLMs (BERT, RoBERTa, DistilBERT) to classify outdated content, achieving $\sim$85% accuracy.
- Benchmarked LLM performance against Scikit-learn models, validating improved precision and recall.

Research

Performance, Patterns, and Path from SQL to NoSQL (Published Poster) | *Sept 2025 – Dec 2025*

Presented: Dec 12, 2025 at Ontario DataBase Day (OnDBD 2025), Western University

- Designed experiments to compare row-store and column-store MySQL with MongoDB across multiple workload types.
- Measured OLTP, OLAP, and pattern-mining performance to analyze throughput, latency, and indexing effects.
- Developed migration workflows from relational schemas to MongoDB to study document structure impacts.
- Wrote SQL and Python scripts to automate tests and collect reproducible performance metrics.
- Examined dataset layouts and access patterns to evaluate architectural trade-offs under heavy workloads.
- Co-authored and presented the research poster to faculty, researchers, and industry experts at OnDBD 2025.

Certifications

NLP with Python for Machine Learning Essential Training, LinkedIn Learning | *March 2025*
Excel with Copilot: AI-Driven Data Analysis, LinkedIn Learning | *Dec 2024*
Linux System Engineer: Networking and SSH, LinkedIn Learning | *Oct 2024*
Microsoft 365 Fundamentals Specialization, Coursera.org | *Dec 2022*